SYSCCA 2.0

Webchat - Installation Guide

VERSION 1.0.0

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1. Pendahuluan

Dokumen ini dibuat untuk memberikan panduan komprehensif mengenai bagaimana mempersiapkan, menguji, dan melakukan instalasi serta *deployment* aplikasi webchat.

2. Prasyarat Umum Pengetahuan

Untuk dapat mengikuti panduan instalasi dan *setup* sistem webchat ini dengan baik, Anda diwajibkan harus familier dan mengerti tentang hal-hal berikut:

- Linux OS
- Sistem Jaringan

3. Production Deployment

3.1. Prasyarat

- Docker (v20.x)
- Docker Compose (v2.x)
- Git
- Nginx
- OpenSSL
- Google Maps API Key (Maps JavaScript API; Maps Static API; Geocoding API;)

3.2. Instalasi Docker

Login ke dalam server/vps anda:

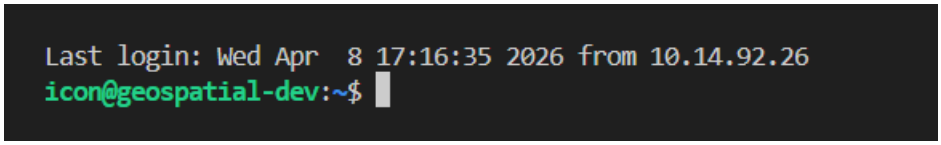
```
ssh icon@10.14.151.119

# Masukkan password akun jika diminta password.
```

```
PS C:\Users\MxK\Desktop\Kiero Project\App\map-batas-wilayah\reactJS-webapp> ssh icon@10.14.151.119
icon@10.14.151.119's password:
```

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Tampilan jika berhasil login:



Pastikan sistem Anda mutakhir dan instal paket bantuan:

- sudo apt update && sudo apt upgrade -y
- sudo apt install -y ca-certificates curl gnupg lsb-release

Masukkan password jika diminta

Cek docker dengan perintah:

- docker --version
- docker compose version

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install docker dengan perintah:

Tambahkan GPG Key resmi

- sudo install -m 0755 -d /etc/apt/keyrings
- curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
- sudo chmod a+r /etc/apt/keyrings/docker.gpg

Daftarkan repositori

- echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

Instalasi Docker & Compose Plugin

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- sudo apt update
- sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

Pengaturan Hak Akses
Ganti 'icon' dengan username Anda jika berbeda

- sudo usermod -aG docker icon

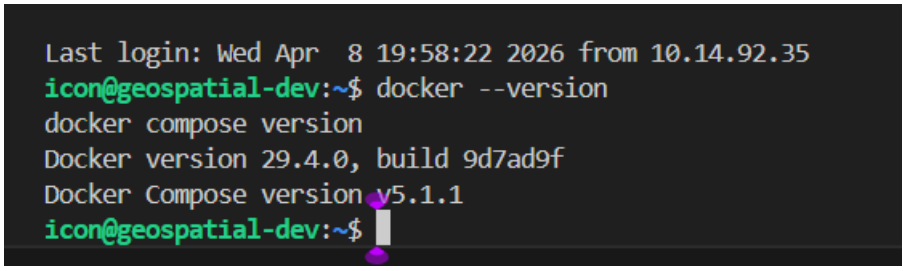
(PENTING: Anda harus logout dan login kembali agar perubahan ini berlaku)

Aktivasi & Verifikasi docker:

- sudo systemctl enable docker
- sudo systemctl start docker

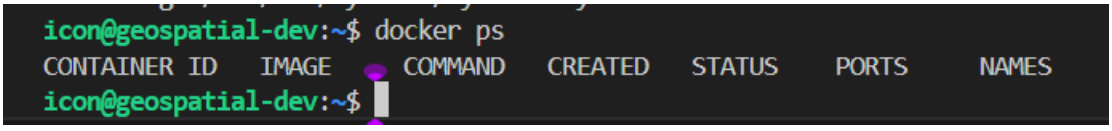
Cek apakah sudah terinstal dengan benar

- docker --version
- docker compose version



Verifikasi Izin Docker:

- docker ps



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3.3. Instalasi Nginx

Cek nginx dengan perintah:

```
nginx -v
systemctl status nginx
```

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install nginx dengan perintah:

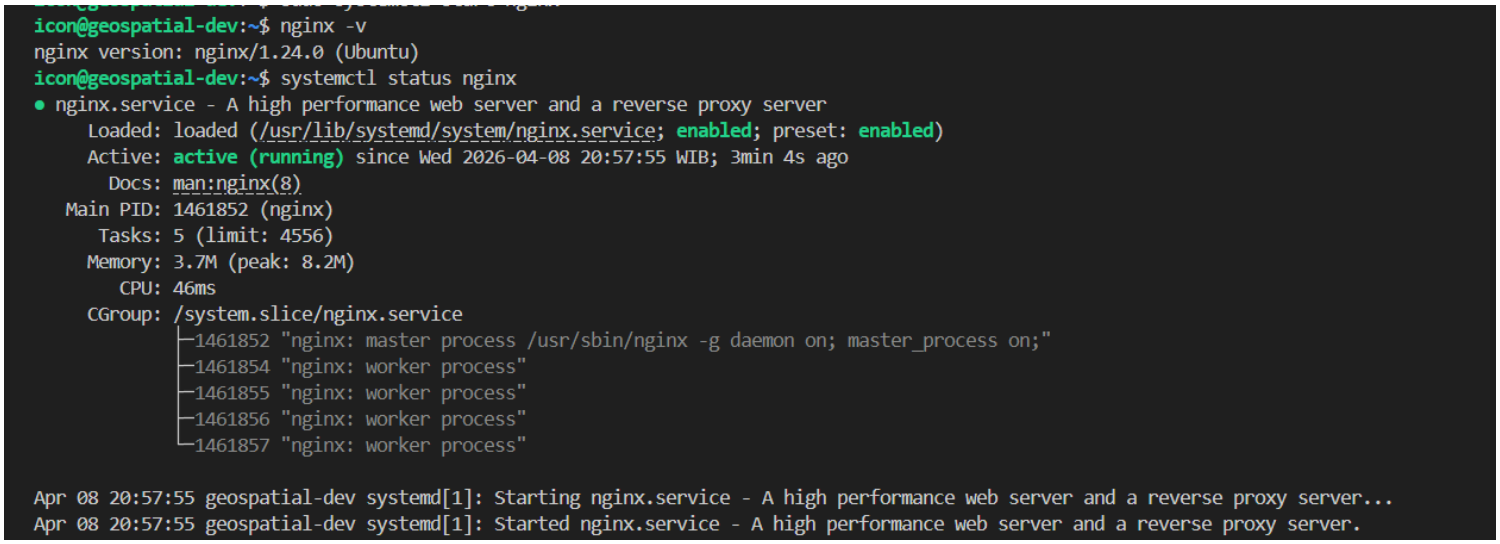
```
sudo apt update

# Instal Nginx
sudo apt install -y nginx

# Pastikan layanan berjalan
sudo systemctl enable nginx
sudo systemctl start nginx
```

Verifikasi instalasi nginx dengan perintah:

```
nginx -v
systemctl status nginx
```



Optimasi File nginx.conf:

```
# Backup konfigurasi nginx yang ada
sudo cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.bak

# Buat SSL DH Param
sudo openssl dhparam -out /etc/nginx/dhparam.pem 2048

# Update nginx.conf dengan konfigurasi ini
sudo bash -c 'cat << EOF > /etc/nginx/nginx.conf
user www-data;
worker_processes auto;
worker_rlimit_nofile 65535;
pid /run/nginx.pid;
include /etc/nginx/modules-enabled/*.conf;

events {
    worker_connections 4096;
    multi_accept on;
    use epoll;
}

http {
```

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```
    # ——— Security Hardening ———————————————————————————————————————
    server_tokens off;
    add_header X-Content-Type-Options "nosniff" always;
    charset utf-8;

    # ——— MIME & Optimization ———————————————————————————————————————
    include      /etc/nginx/mime.types;
    default_type  application/octet-stream;

    # ——— Buffer Optimization ———————————————————————————————————————
    client_body_buffer_size    16K;
    client_header_buffer_size  1k;
    client_max_body_size       50M;
    large_client_header_buffers 4 8k;

    # ——— Timeouts ——————————————————————————————————————————————————
    client_body_timeout   12;
    client_header_timeout 12;
    keepalive_timeout     65;
    send_timeout          10;

    # ——— SSL Hardening (Mozilla Intermediate) ——————————————————————
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;
    ssl_session_cache   shared:SSL:10m;
    ssl_session_timeout 1d;
    ssl_session_tickets off;

    # DH Parameters untuk Forward Secrecy
    ssl_dhparam /etc/nginx/dhparam.pem;

    # ——— Gzip Compression ——————————————————————————————————————————
```

```
    gzip on;
    gzip_vary on;
    gzip_proxied any;
    gzip_comp_level 6;
    gzip_min_length 256;
    gzip_types
        text/plain
        text/css
        text/xml
        text/javascript
        application/json
        application/javascript
        application/xml
        image/svg+xml
        font/woff2;

    # ─── Logging ──────────────────────────────────────────────────────
──────────────────────────────────

    access_log /var/log/nginx/access.log;
    error_log  /var/log/nginx/error.log;

    # ─── Includes ─────────────────────────────────────────────────────
──────────────────────────────────

    include /etc/nginx/conf.d/*.conf;
    include /etc/nginx/sites-enabled/*;
}
EOF'
```

Optimasi konfigurasi proxy param. Buka dan edit proxy_params:

```
sudo nano /etc/nginx/proxy_params
```

Isi proxy_params dengan konfigurasi berikut:

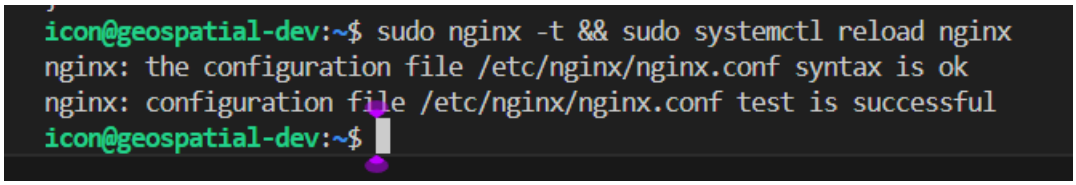
```
proxy_set_header Host            $http_host;
proxy_set_header X-Real-IP       $remote_addr;
proxy_set_header X-Forwarded-For   $proxy_add_x_forwarded_for;
```

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```
proxy_set_header X-Forwarded-Proto $scheme;
proxy_set_header X-Forwarded-Host  $host;
proxy_set_header X-Forwarded-Port  $server_port;
```

Tes dan reload konfigurasi:

```
sudo nginx -t && sudo systemctl reload nginx
```



3.4. Instalasi Git & OpenSSL

Cek Git & OpenSSL dengan perintah:

```
# Cek Git
git --version

# Cek OpenSSL
openssl version
```

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install nginx dengan perintah:

```
sudo apt install -y git openssl
```

Verifikasi instalasi Git & OpenSSL dengan perintah:

```
nginx -v
systemctl status nginx
```

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```
git version 2.43.0
OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
icon@geospatial-dev:~$
```

3.5. Konfigurasi Firewall

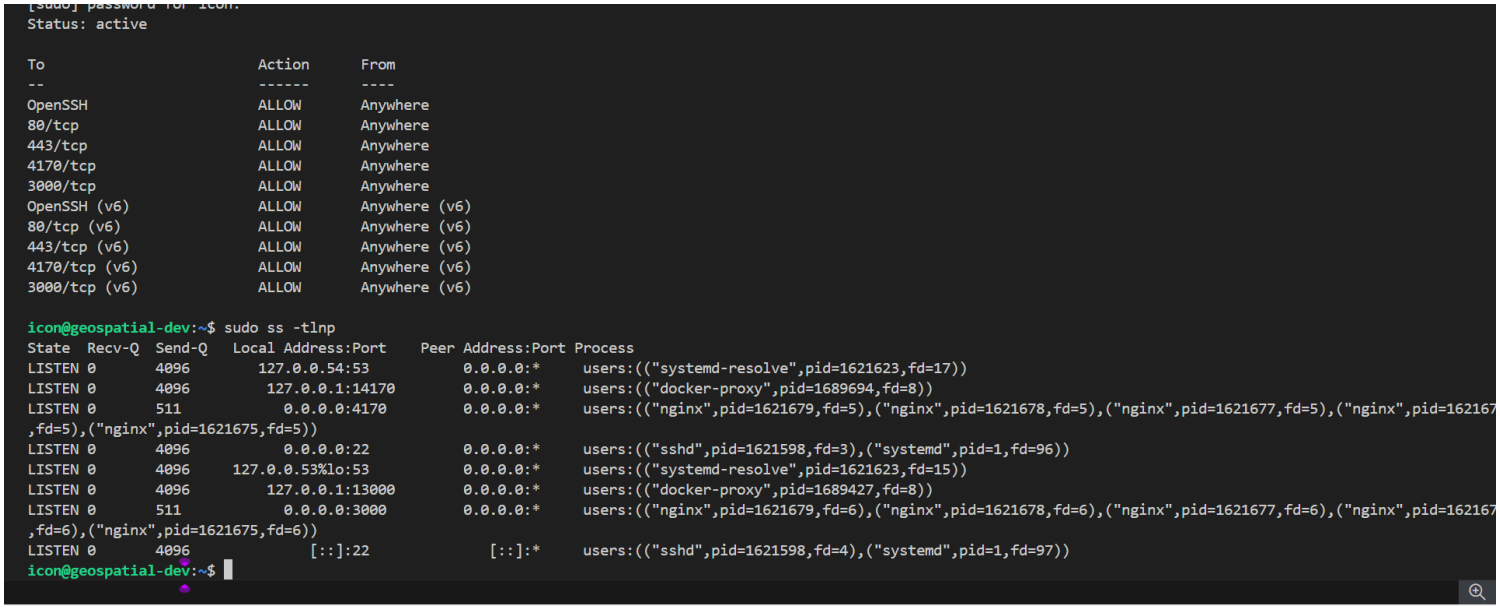
Cek status firewall:

Cek firewall

- sudo ufw status

Cek port yang sedang digunakan

- sudo ss -tlnp



```
[sudo] password for icon:
Status: active

To                         Action      From
--                         ------      ----
OpenSSH                    ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
4170/tcp                   ALLOW       Anywhere
3000/tcp                   ALLOW       Anywhere
OpenSSH (v6)               ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
4170/tcp (v6)              ALLOW       Anywhere (v6)
3000/tcp (v6)              ALLOW       Anywhere (v6)

icon@geospatial-dev:~$ sudo ss -tlnp
State  Recv-Q  Send-Q   Local Address:Port    Peer Address:Port Process
LISTEN 0       4096        127.0.0.54:53          0.0.0.0:*     users:(("systemd-resolve",pid=1621623,fd=17))
LISTEN 0       4096         127.0.0.1:14170       0.0.0.0:*     users:(("docker-proxy",pid=1689694,fd=8))
LISTEN 0       511            0.0.0.0:4170        0.0.0.0:*     users:(("nginx",pid=1621679,fd=5),("nginx",pid=1621678,fd=5),("nginx",pid=1621677,fd=5),("nginx",pid=162167
,fd=5),("nginx",pid=1621675,fd=5))
LISTEN 0       4096           0.0.0.0:22          0.0.0.0:*     users:(("sshd",pid=1621598,fd=3),("systemd",pid=1,fd=96))
LISTEN 0       4096     127.0.0.53%lo:53          0.0.0.0:*     users:(("systemd-resolve",pid=1621623,fd=15))
LISTEN 0       4096        127.0.0.1:13000       0.0.0.0:*     users:(("docker-proxy",pid=1689427,fd=8))
LISTEN 0       511            0.0.0.0:3000        0.0.0.0:*     users:(("nginx",pid=1621679,fd=6),("nginx",pid=1621678,fd=6),("nginx",pid=1621677,fd=6),("nginx",pid=162167
,fd=6),("nginx",pid=1621675,fd=6))
LISTEN 0       4096             [::]:22             [::]:*     users:(("sshd",pid=1621598,fd=4),("systemd",pid=1,fd=97))
icon@geospatial-dev:~$
```

Lihat Local Address:Port. Daftarkan semua port kecuali yang localhost(127.0.0.x):

1. Izinkan SSH Jika belum (WAJIB PERTAMA, agar tidak terkunci dari server!)

- sudo ufw allow OpenSSH

2. Izinkan rute Web jika belum (untuk Nginx Reverse Proxy)

- sudo ufw allow 80/tcp
- sudo ufw allow 443/tcp

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3. Izinkan port publik aplikasi Webchat

- sudo ufw allow 4171/tcp # Widget CDN
- sudo ufw allow 4172/tcp # Agent Dashboard
- sudo ufw allow 3002/tcp # Widget API
- sudo ufw allow 3003/tcp # Agent API

Aktifkan firewal dan verifikasi

Aktifkan Firewall Jika belum (ketik 'y' saat diminta konfirmasi)

- sudo ufw enable

Verifikasi status firewall sudah aktif dan port sudah terdaftar

- sudo ufw status

```
Status: active

To                         Action      From
--                         ------      ----
OpenSSH                    ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
4170/tcp                   ALLOW       Anywhere
3000/tcp                   ALLOW       Anywhere
4171/tcp                   ALLOW       Anywhere
4172/tcp                   ALLOW       Anywhere
3002/tcp                   ALLOW       Anywhere
3003/tcp                   ALLOW       Anywhere
OpenSSH (v6)               ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
4170/tcp (v6)              ALLOW       Anywhere (v6)
3000/tcp (v6)              ALLOW       Anywhere (v6)
4171/tcp (v6)              ALLOW       Anywhere (v6)
4172/tcp (v6)              ALLOW       Anywhere (v6)
3002/tcp (v6)              ALLOW       Anywhere (v6)
3003/tcp (v6)              ALLOW       Anywhere (v6)
```

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3.6. Kloning Repositori

Buat SSH key khusus proyek Webchat. Kita akan menggunakan algoritma ed25519.

Jalankan perintah:

```
ssh-keygen -t ed25519 -C "webchat-deploy-key" -f ~/.ssh/id_ed25519_webchat -N ""
```



Daftarkan "Alias" Khusus di SSH Config dengan perintah:

```
# Buat file config jika belum ada
touch ~/.ssh/config

# Tambahkan konfigurasi host alias
cat << 'EOF' >> ~/.ssh/config
Host github.com-webchat
 HostName github.com
 User git
 IdentityFile ~/.ssh/id_ed25519_webchat
 IdentitiesOnly yes
EOF
```

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Verifikasi konfigurasi SSH:

```
cat ~/.ssh/config
```

```
icon@geospatial-dev:~$ cat ./.ssh/config
Host github.com-geomap
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519_geomap
  IdentitiesOnly yes
Host github.com-webchat
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519_webchat
  IdentitiesOnly yes
icon@geospatial-dev:~$
```

Ambil public key untuk github:

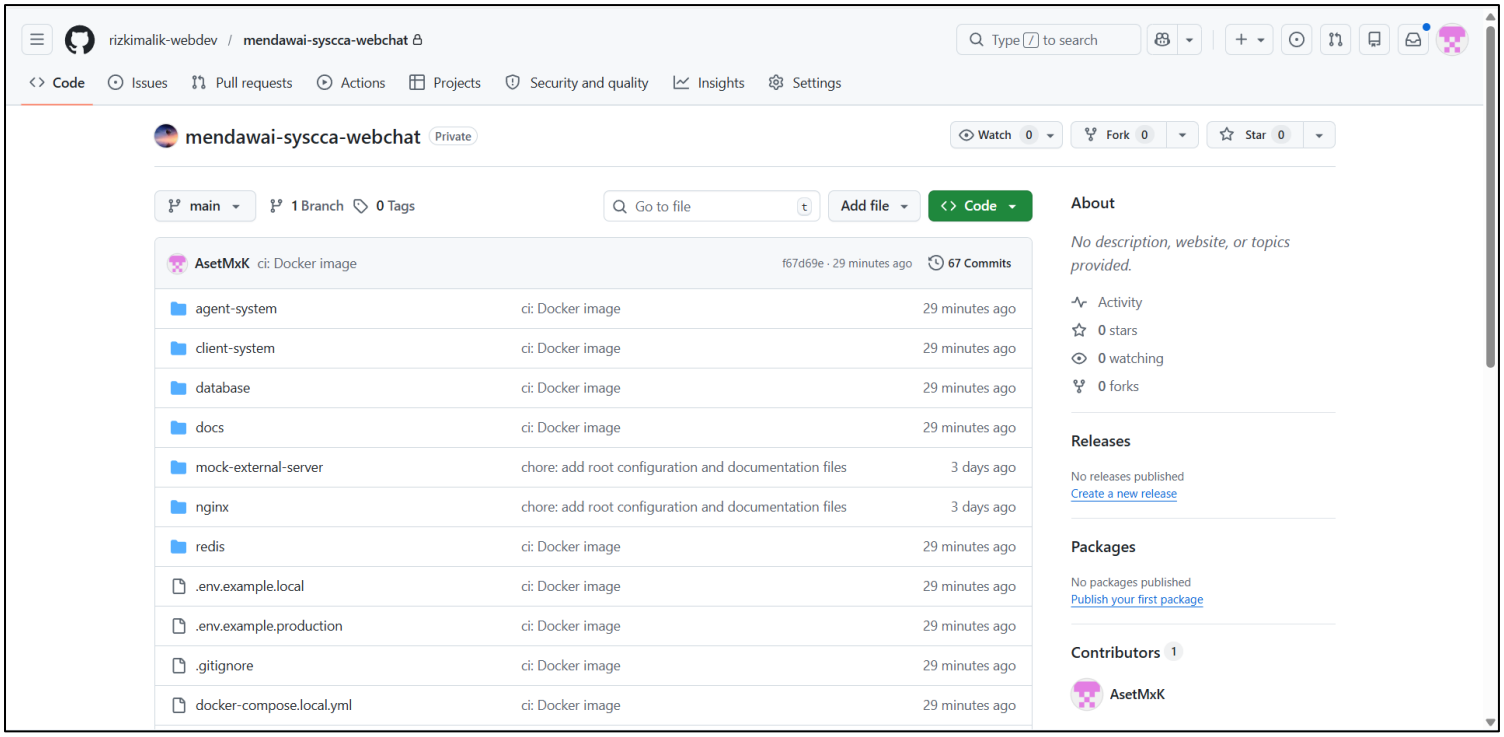
```
cat ~/.ssh/id_ed25519_webchat.pub

# Salin dan simpan output-nya (dimulai dari ssh-ed25519)
```

Daftarkan di GitHub Repository:

1. Buka browser komputer Anda dan akses **Repository GitHub** webchat Anda.

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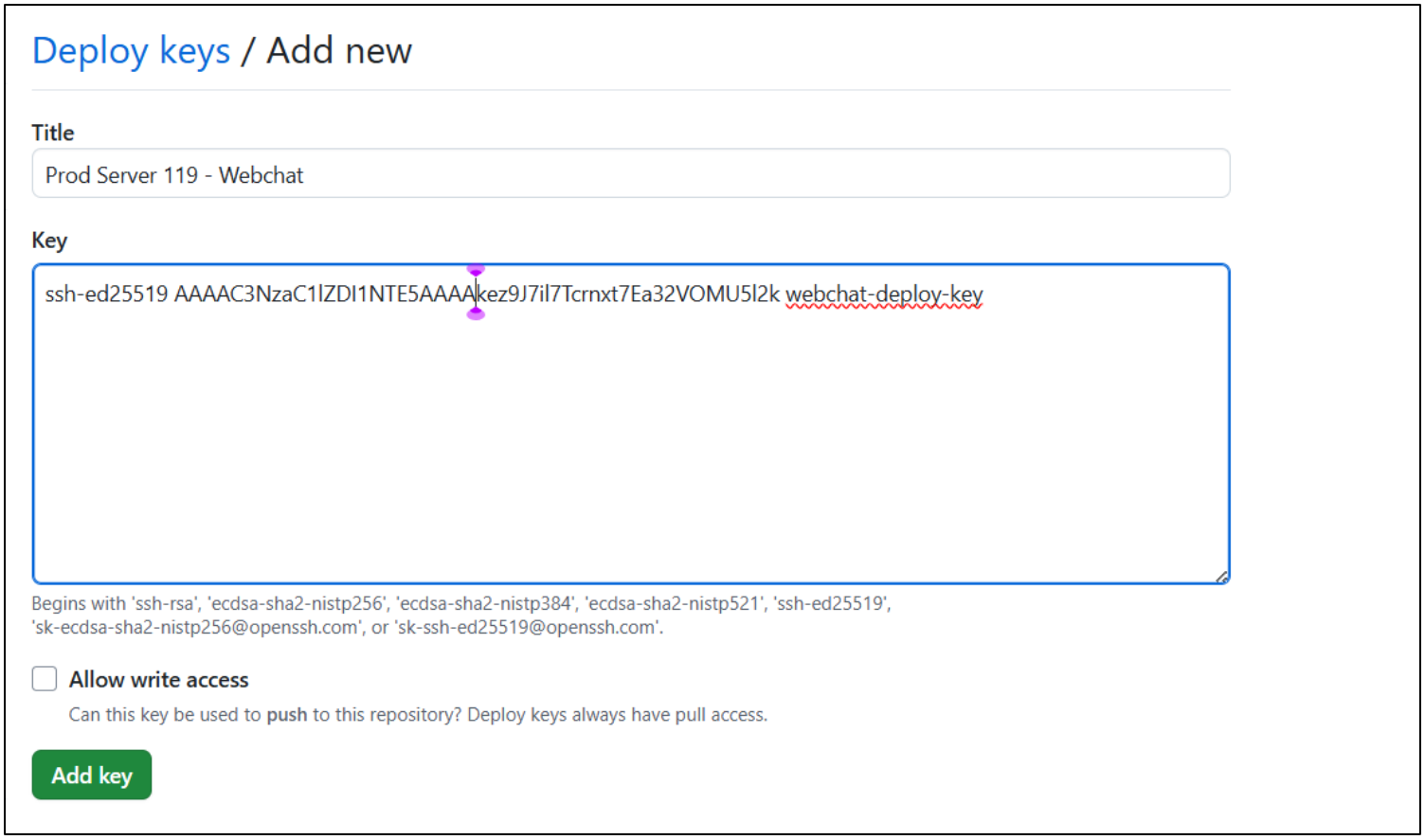


2. Klik tab **Settings** (Pengaturan repositori).
3. Di panel kiri, masuk ke menu **Deploy keys** (dibawah bagian *Security*).

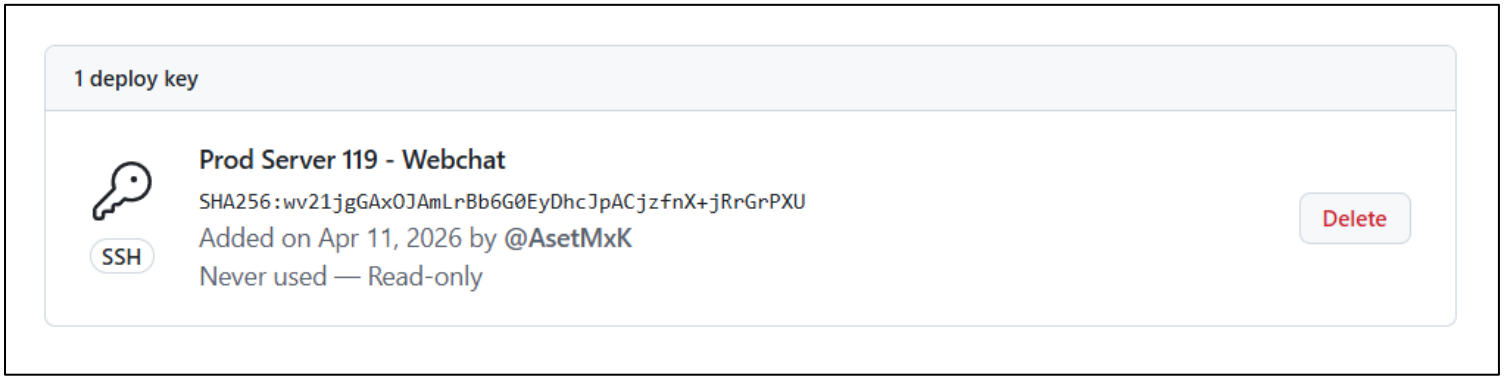


4. Klik tombol **Add deploy key**.

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5. Isi formulir:
    - **Title**: Prod Server 119 - Webchat
    - **Key**: *Paste* (tempelkan) *Public Key* webchat di sini.
    - **Allow write access**: **JANGAN DICENTANG** (Biarkan Read-only demi keamanan server).
6. Klik **Add key**.



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Uji koneksi:

- ssh -T git@github.com-webchat

```
icon@geospatial-dev:~$ ssh -T git@github.com-webchat
Hi AsetMxK/Webchat! You've successfully authenticated, but GitHub does not provide shell access.
icon@geospatial-dev:~$
```

Kloning repositori:

- cd /home/icon

PASTIKAN GANTI TULISAN USERNAME DI BAWAH DENGAN AKUN GITHUB ANDA DAN REPOSITORI DENGAN NAMA REPOSITORI ANDA

- git clone git@github.com-webchat:USERNAME/REPOSITORI.git webchat

```
icon@geospatial-dev:~$ git clone git@github.com-webchat:AsetMxK/Webchat.git webchat
Cloning into 'webchat'...
remote: Enumerating objects: 1923, done.
remote: Counting objects: 100% (1923/1923), done.
remote: Compressing objects: 100% (769/769), done.
remote: Total 1923 (delta 1064), reused 1923 (delta 1064), pack-reused 0 (from 0)
Receiving objects: 100% (1923/1923), 1.81 MiB | 1.41 MiB/s, done.
Resolving deltas: 100% (1064/1064), done.
icon@geospatial-dev:~$
```

3.7. Setup proyek

Buat folder keys & logs:

- cd /home/icon/webchat

Pastikan folder keys dan logs tersedia

- mkdir -p keys
- mkdir -p logs

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Membuat SSL (selfsigned) untuk https:

```
cat << 'EOF' > generate_ssl.sh
#!/bin/bash
# 🔐 Optimized Self-Signed SSL Generator with SAN support

# 1. Deteksi IP Otomatis atau Gunakan Argumen
# Jika menjalankan ./generate_ssl.sh 1.2.3.4 maka akan pakai IP tersebut.
# Jika tidak, akan mendeteksi IP utama server.
DEFAULT_IP=$(hostname -I | awk '{print $1}')
SERVER_IP=${1:-$DEFAULT_IP}

KEY_DIR="./keys/ssl"
CONFIG_FILE="./keys/ssl/openssl.cnf"
mkdir -p $KEY_DIR

echo "🔍 Target IP: $SERVER_IP"

# 2. Buat file konfigurasi OpenSSL sementara
cat << EOC > $CONFIG_FILE
[req]
default_bits       = 2048
distinguished_name = req_distinguished_name
x509_extensions    = v3_req
prompt             = no

[req_distinguished_name]
C  = ID
ST = DKI Jakarta
L  = Jakarta
O  = Kiero Project
OU = IT Department
CN = $SERVER_IP

[v3_req]
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
extendedKeyUsage = serverAuth
subjectAltName = @alt_names
```

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```
[alt_names]
IP.1 = $SERVER_IP
IP.2 = 127.0.0.1
DNS.1 = localhost
EOC

echo "🔐 Generating SSL Certificate (Valid for 10 years)..."

openssl req -x509 -nodes -days 3650 -newkey rsa:2048 \
   -keyout "$KEY_DIR/nginx.key" \
   -out "$KEY_DIR/nginx.crt" \
   -config $CONFIG_FILE

# Hapus file konf sementara
rm $CONFIG_FILE

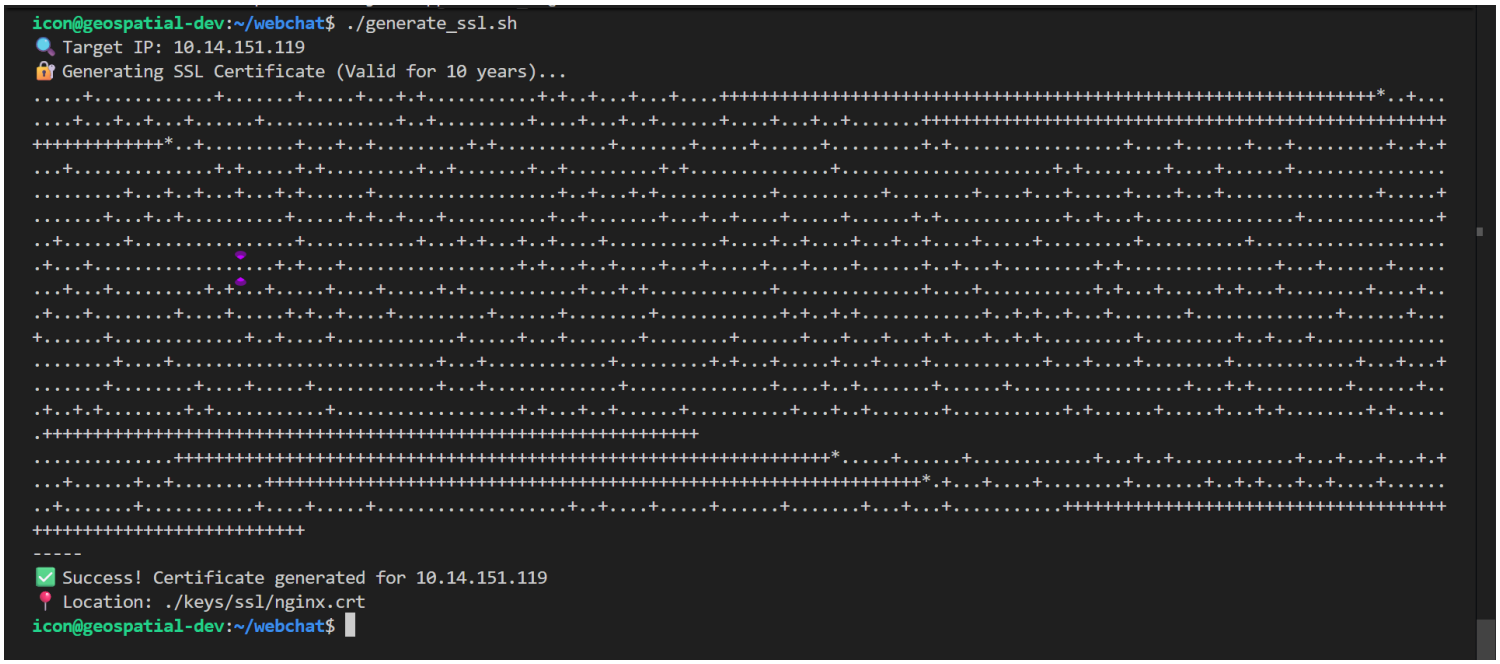
echo "☑ Success! Certificate generated for $SERVER_IP"
echo "📍 Location: $KEY_DIR/nginx.crt"
EOF
```

Jalankan skrip:

```
# Berikan izin eksekusi
chmod +x generate_ssl.sh

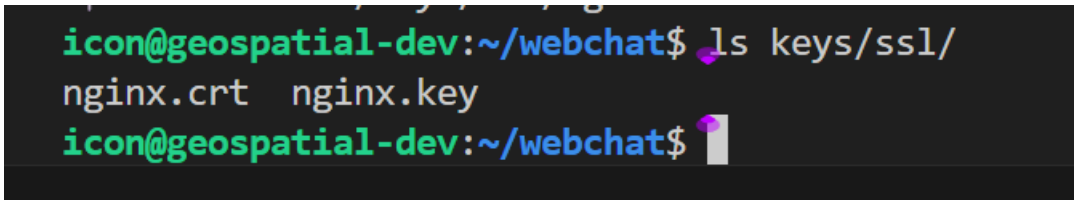
# Eksekusi pembuat sertifikat 10 tahun
./generate_ssl.sh
```

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Verifikasi sertifikat SSL:

```
ls keys/ssl/
```



Salin SSL ke direktori host Nginx:

```
sudo cp keys/ssl/nginx.crt /etc/ssl/certs/webchat-selfsigned.crt
sudo cp keys/ssl/nginx.key /etc/ssl/private/webchat-selfsigned.key
```

Membuat auto-renewal SSL:

```
cat << 'EOF' > renew_ssl.sh
#!/bin/bash
# 🔄 SSL Auto-Renewal Script for Self-Signed Certs
```

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```
PROJECT_DIR="/home/icon/webchat"
CERT_FILE="$PROJECT_DIR/keys/ssl/nginx.crt"
DAYS_BEFORE_EXPIRY=30

echo "🧮 Checking SSL certificate expiry..."

# 1. Cek apakah file sertifikat ada
if [ ! -f "$CERT_FILE" ]; then
   echo "❌ Certificate not found at $CERT_FILE. Generating new one..."
   cd $PROJECT_DIR && ./generate_ssl.sh
else
   # 2. Cek sisa umur sertifikat dalam detik
   # openssl x509 -checkend akan return 0 jika cert masih valid sampai N detik ke depan
   EXPIRY_SECONDS=$((DAYS_BEFORE_EXPIRY * 24 * 3600))

   if openssl x509 -checkend $EXPIRY_SECONDS -noout -in "$CERT_FILE"; then
      echo "☑️ Certificate is still valid for more than $DAYS_BEFORE_EXPIRY days. No renewal needed."
      exit 0
   else
      echo "⚠️ Certificate is expiring soon (less than $DAYS_BEFORE_EXPIRY days). Renewing..."
      cd $PROJECT_DIR && ./generate_ssl.sh
   fi
fi

# 3. Salin ke direktori sistem Nginx (Requires sudo)
echo "🗁 Copying new certificates to system directory..."
sudo cp "$PROJECT_DIR/keys/ssl/nginx.crt" /etc/ssl/certs/webchat-selfsigned.crt
sudo cp "$PROJECT_DIR/keys/ssl/nginx.key" /etc/ssl/private/webchat-selfsigned.key

# 4. Reload Nginx
echo "🚀 Reloading Nginx..."
sudo systemctl reload nginx
```

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```
echo "✨ Renewal process completed successfully."
EOF
```

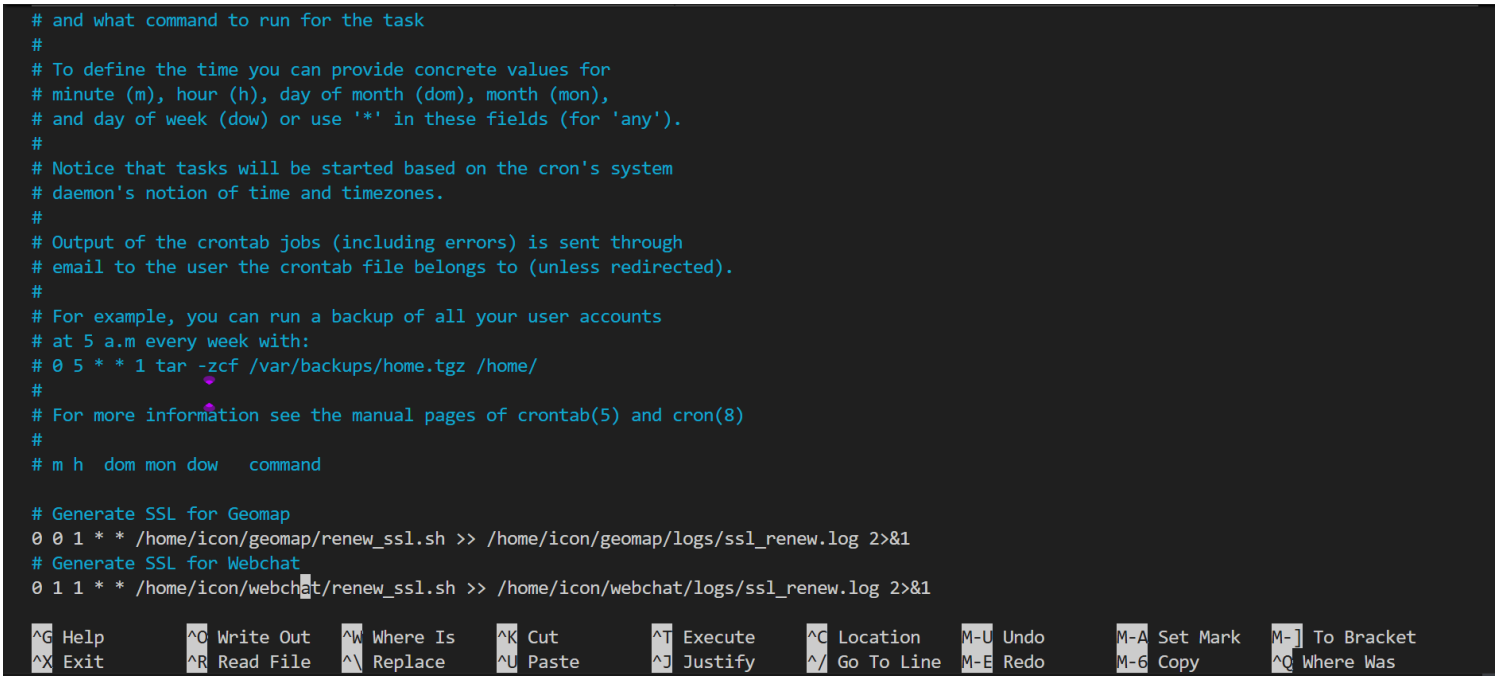
Setup auto-renewal dengan cron job:

```
# Berikan izin eksekusi skrip auto-renewal SSL
chmod +x renew_ssl.sh

# Buka editor crontab
crontab -e
```

Tambahkan baris ini di paling bawah:

```
# menjalankan pengecekan setiap tanggal 1 jam 01:00
0 1 1 * * /home/icon/webchat/renew_ssl.sh >> /home/icon/webchat/logs/ssl_renew.log 2>&1
```



Konfigurasi .env:

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Salin .env.example.production

- cd /home/icon/webchat
- cp .env.example.production .env.production
- cp agent-system/apps/backend/.env.example.production agent-system/apps/backend/.env.production
- cp client-system/packages/widget-backend/.env.example.production client-system/packages/widget-backend/.env.production
- cp agent-system/apps/frontend/.env.example.production agent-system/apps/frontend/.env.production
- cp client-system/packages/widget-frontend/.env.example.production client-system/packages/widget-frontend/.env.production

3.8. Konfigurasi .env.production agent-backend

Buka dan edit .env.production agent-backend

- nano agent-system/apps/backend/.env.production

Contoh isi:

- NODE_ENV=production
- PORT=3003
- LOG_LEVEL=info

Database (Production URL - uses internal Docker network hostname)

- DATABASE_URL="postgresql://postgres:e7051a15a18f182b2e0b49df8e3bc256@postgres:5432/agent_db?schema=public"

Redis (Production URL - uses internal Docker network hostname)

- REDIS_URL="redis://:48b70efd02a6000e7bfa542ddd7651e6@redis:6379"

Security (Generated secure keys)

- JWT_SECRET="dbe0a6db196f3d0d1f60b45ed9f6d99069b8cabb1707b5a4ca44ae2cdb309d89"

- WEBHOOK_SECRET="b5774a622c55842816133d4ef07dffadadb6aa2f67825ed52d52b53f12717813"
- ENCRYPTION_KEY="72248c7283780e39c530aa0923dc60b4e6b0f8f53f9081b1f50ea7a5b7f431fc"

Frontend URL (Used for CORS - pointing to the Agent Dashboard port)

- FRONTEND_URL="https://10.14.151.119:4172"

Rate Limiting

- RATE_LIMIT_LOGIN_MAX=20
- RATE_LIMIT_REFRESH_MAX=50
- RATE_LIMIT_UPLOAD_MAX=20
- RATE_LIMIT_WIDGET_MAX=300
- RATE_LIMIT_WINDOW="1 minute"
- RATE_LIMIT_ALLOW_LIST="127.0.0.1"

Token Expiry

- JWT_ACCESS_EXPIRY="15m"
- JWT_REFRESH_EXPIRY="7d"

Anda bisa melihat perbedaannya DI SINI

3.9. Konfigurasi .env.production widget-backend

Buka dan edit .env.production widget-backend

- nano client-system/packages/widget-backend/.env.production

Contoh isi:

- NODE_ENV=production
- PORT=3002
- LOG_LEVEL=info

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```
# Redis Connection (Uses internal Docker network)
REDIS_URL="redis://:48b70efd02a6000e7bfa542ddd7651e6@redis:6379"

# Security (Webhook secret must match Agent Backend)
WEBHOOK_TIMEOUT=5000
WEBHOOK_MAX_RETRIES=3

# WEBHOOK_SECRET harus sama dengan yang di agent-backend
WEBHOOK_SECRET="b5774a622c55842816133d4ef07dffadadb6aa2f67825ed52d52b53f12717813"

# Internal connection to Agent Backend (within Docker network)
AGENT_BACKEND_URL="http://agent-backend:3003"

# Public URL of Agent Backend for file downloads/attachments
AGENT_BACKEND_PUBLIC_URL="https://10.14.151.119:3003"
```

Anda bisa melihat perbedaannya DI SINI

3.10. Konfigurasi .env.production agent-frontend

```
# Buka dan edit .env.production agent-frontend
nano agent-system/apps/frontend/.env.production

# Contoh isi:
# Connects to the Global Nginx Proxy (which routes to agent-backend 3003)
VITE_API_URL=https://10.14.151.119:3003/api

# Base path if proxy maps domain.com/webchat/ to the frontend:
VITE_BASE_PATH=/
VITE_SOCKET_PATH=

# Google Maps API Key
VITE_GOOGLE_MAPS_API_KEY=AIzaSy-ContohAPIKeyGoogleMapsDev000
```

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```
# Connects to the Widget Frontend
VITE_WIDGET_FRONTEND_URL=https://10.14.151.119:4171
```

Anda bisa melihat perbedaannya DI SINI

3.11. Konfigurasi .env.production widget-frontend

```
# Buka dan edit .env.production widget-frontend
nano client-system/packages/widget-frontend/.env.production

# Contoh isi:
# Public API URL for the Widget Backend
VITE_WIDGET_BACKEND_URL=https://10.14.151.119:3002
VITE_BASE_PATH=/
VITE_SOCKET_PATH=

# Google Maps API Key
VITE_GOOGLE_MAPS_API_KEY=AIzaSy-ContohAPIKeyGoogleMapsDev000

# Dev Client ID (Can be empty in real production unless needed)
VITE_DEV_CLIENT_ID=
```

Anda bisa melihat perbedaannya DI SINI

3.12. Konfigurasi .env.production webchat

```
# Buka dan edit .env.production
nano .env.production

# Contoh isi:
POSTGRES_DB=agent_db
```

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- POSTGRES_USER=postgres
- POSTGRES_PASSWORD=e7051a15a18f182b2e0b49df8e3bc256

- REDIS_PASSWORD=48b70efd02a6000e7bfa542ddd7651e6

Container Port Mappings (Exposed to Host 127.0.0.1)

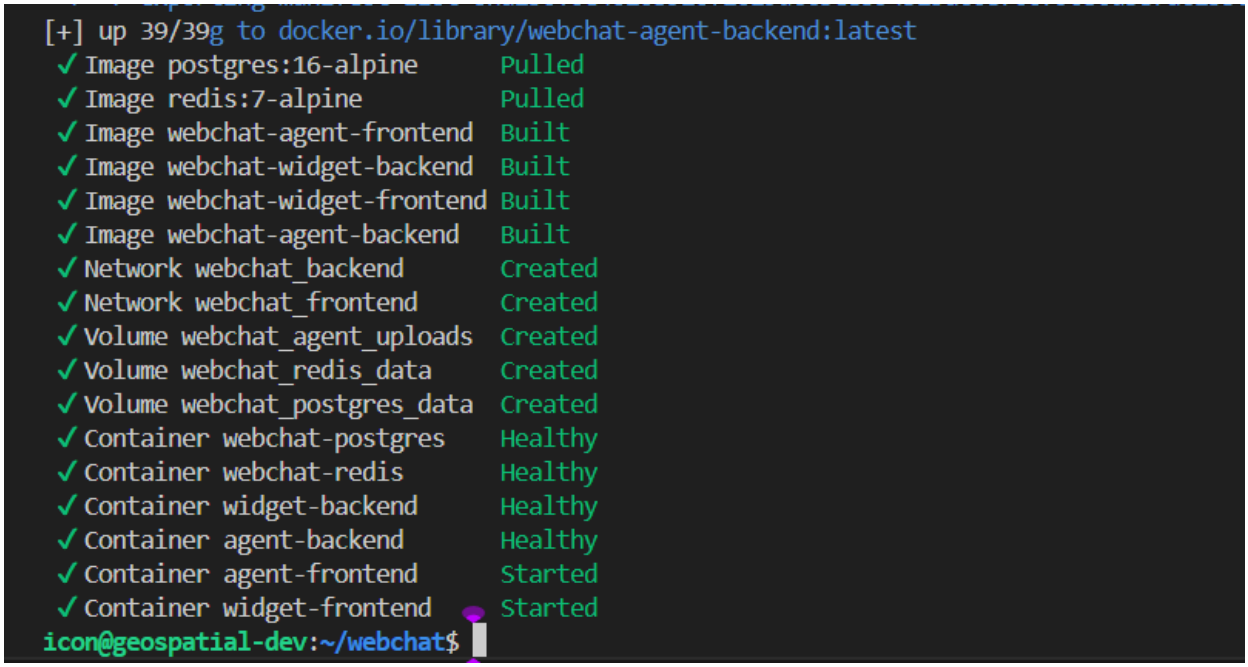
- AGENT_FRONTEND_PORT=14172
- AGENT_BACKEND_PORT=13003
- WIDGET_FRONTEND_PORT=14171
- WIDGET_BACKEND_PORT=13002

Anda bisa melihat perbedaannya DI SINI

3.13. Jalankan deployment

Jalankan perintah:

- docker compose --env-file .env.production -f docker-compose.prod.yml up -d --build



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3.14. Migrasi database dan seeding

Jalankan perintah:

- docker compose -f docker-compose.prod.yml exec agent-backend npx prisma migrate deploy
- docker compose -f docker-compose.prod.yml exec agent-backend npx prisma db seed

```
icon@geospatial-dev:~/webchat$ docker compose -f docker-compose.prod.yml exec agent-backend npx prisma migrate deploy
[dotenv@17.2.3] injecting env (0) from .env -- tip: backup and recover secrets: https://dotenvx.com/ops
Loaded Prisma config from prisma.config.js.

Prisma schema loaded from prisma/schema.prisma.
Datasource "db": PostgreSQL database "agent_db", schema "public" at "postgres:5432"

6 migrations found in prisma/migrations


No pending migrations to apply.
icon@geospatial-dev:~/webchat$ docker compose -f docker-compose.prod.yml exec agent-backend npx prisma db seed
[dotenv@17.2.3] injecting env (0) from .env -- tip: audit secrets and track compliance: https://dotenvx.com/ops
Loaded Prisma config from prisma.config.js.

Running seed command `node prisma/seed.js` ...
[dotenv@17.2.3] injecting env (0) from .env -- tip: enable debug logging with { debug: true }
Seeding database...
Seeding finished.

The seed command has been executed.
icon@geospatial-dev:~/webchat$
```

3.15. Publish aplikasi ke internet(publik)

Salin dan Aktifkan Konfigurasi Sites-Available. Jalankan perintah:

1. Salin file dari folder proyek ke direktori sistem Nginx

- sudo cp /home/icon/webchat/nginx/webchat.conf /etc/nginx/sites-available/webchat

2. Aktifkan konfigurasi dengan membuat symbolic link

- sudo ln -sf /etc/nginx/sites-available/webchat /etc/nginx/sites-enabled/

3. Hapus konfigurasi default Ubuntu (agar tidak terjadi konflik port)

- sudo rm -f /etc/nginx/sites-enabled/default

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Verifikasi dan muat ulang nginx. Jalankan perintah:

Verifikasi konfigurasi nginx

- sudo nginx -t

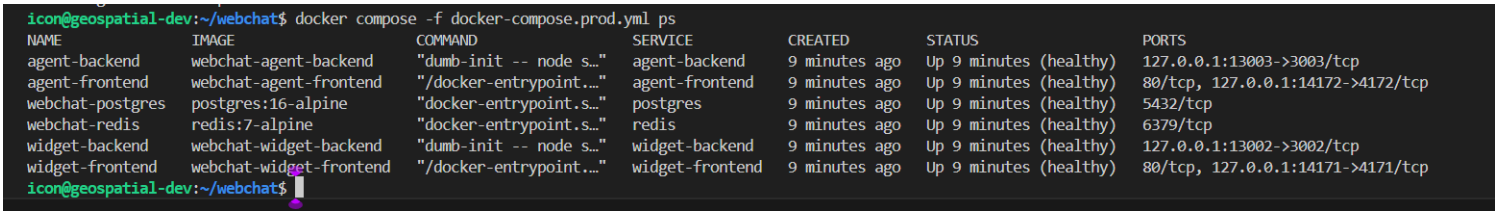
Muat ulang nginx

- sudo systemctl reload nginx

3.16. Verifikasi

Jalankan perintah:

- docker compose -f docker-compose.prod.yml ps



Akses:

Agent Frontend: https://10.14.151.119:4172

widget: https://10.14.151.119:4171/embed.iife.js

4. Maintenance

4.1. Pengecekan dan Pembaruan Manual SSL

Jalankan perintah:

- cd /home/icon/webchat
- ./renew_ssl.sh

4.2. Pengecekan Status Layanan

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Pengecekan Layanan Nginx:

```
sudo systemctl status nginx
```

Pengecekan Layanan Docker:

```
sudo systemctl status docker
```

Pengecekan Layanan Webchat:

```
cd /home/icon/webchat
docker compose -f docker-compose.prod.yml ps
```

4.3. Monitoring proses layanan

Monitoring Layanan Webchat:

```
cd /home/icon/webchat
docker compose -f docker-compose.prod.yml stats
```

4.4. Monitoring logs

Log Webchat:

```
cd /home/icon/webchat
docker compose -f docker-compose.prod.yml logs -f

# Jika ingin melihat layanan tertentu
# docker compose -f docker-compose.prod.yml logs -f agent-backend
```

5. Troubleshooting

5.1. DNS Hijacking & Kegagalan Resolusi Nama

- **Masalah**: Muncul pesan Temporary failure in name resolution atau 404 Not Found saat menjalankan perintah apt update / apt install.
- **Penyebab**: Jaringan korporat atau ISP melakukan *DNS Hijacking* yang membelokkan permintaan DNS (seperti ke archive.ubuntu.com) menuju IP palsu.
- **Solusi**: Gunakan strategi *Static DNS*. Temukan IP mirror Ubuntu yang masih aktif (mengembalikan status HTTP 200) dengan mem-bypass DNS secara manual, lalu daftarkan IP tersebut langsung ke dalam file /etc/hosts server. Setelah didaftarkan, bersihkan *cache* paket menggunakan perintah sudo apt clean dan sudo rm -rf /var/lib/apt/lists/*, lalu jalankan kembali sudo apt update.

Jalankan perintah:

```
for ip in 185.125.190.36 185.125.190.39 91.189.91.81 91.189.91.82
  91.189.91.83 91.189.91.84; do
 echo -n "Testing $ip... "
 curl -s -o /dev/null -w "HTTP %{http_code}" --connect-timeout 5 \
  "http://$ip/ubuntu/dists/noble/InRelease" -H "Host: archive.ubuntu.com"
 echo ""
done
```

Daftarkan IP yang berhasil (status: 200):

```
echo "91.189.91.82 archive.ubuntu.com" | sudo tee -a /etc/hosts
echo "91.189.91.82 security.ubuntu.com" | sudo tee -a /etc/hosts
```

Hapus cache lama dan update metadata:

```
# Hapus cache
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*
```

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```
# Update metadata dari NOL
sudo apt update
```

5.2. Akses Ditolak pada Docker (Permission Denied)

- **Masalah**: Terjadi *error* dengan pesan permission denied while trying to connect to the Docker daemon socket.
- **Penyebab**: Akun *user* yang Anda gunakan (contoh: icon) belum ditambahkan ke dalam grup keamanan docker.
- **Solusi**: Tambahkan *user* tersebut ke grup docker dengan menjalankan perintah sudo usermod -aG docker icon. **Sangat penting:** Anda wajib untuk *logout* dan *login* kembali ke dalam server agar perubahan grup tersebut diterapkan.

Jalankan perintah:

```
sudo usermod -aG docker icon

# logout dan login ulang
logout
```

5.3. APT Mengembalikan Status 404 Not Found (Metadata Usang)

- **Masalah**: Instalasi gagal dengan *error* `404 Not Found` ketika menggunakan `apt install`, sekalipun koneksi internet dan jaringan server repositori dalam keadaan normal.
- **Penyebab**: File metadata (cache) Ubuntu lokal Anda di `/var/lib/apt/lists/` belum diperbarui dan masih mencari file `.deb` versi lama yang sudah tidak tersedia lagi di server pusat.

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- **Solusi**: Hapus seluruh metadata lokal yang sudah usang dengan menjalankan `sudo apt clean` dan `sudo rm -rf /var/lib/apt/lists/*`. Segera setelah itu, perbarui ulang indeks repositori dengan menjalankan `sudo apt update`, kemudian ulangi instalasi paket Anda.

  Jalankan perintah:

```
# Hapus semua cache metadata lama
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*

# Download metadata segar dari repositori
sudo apt update

# Ulangi instalasi (sekarang merujuk ke versi terbaru)
sudo apt install -y <package-name>
```

6. End of Document

If you encounter problem or difficulties, you may reach our support contact at :

1. WhatsApp or GSM contact to : +62 819 4401 1272
2. Email to support@kieroteknologi.com
3. Visit kieroteknologi.com and post your problem in contact us section

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